

BORAMA HARDWARE

Product Requirements Document

AI-Powered Hardware Retail Platform

Home Depot-Style Store • Borama, Somaliland

Version 1.0 • June 2025 • Confidential

Field	Details
Document Type	Product Requirements Document (PRD)
Project	Borama Hardware Store
Platform	Web Application (PWA)
Version	1.0
Date	June 2025
Status	Final — Ready for Development
Stack	Next.js 14 · Supabase · Vercel · Google Gemini API

Table of Contents

#	Section	Page
1.	Executive Summary	3
2.	Problem Statement	3
3.	Users & Roles	3
4.	Feature Requirements	4
4.1	Customer-Facing Store	4
4.2	Admin / Store Owner	7
4.3	Inventory Manager	9
4.4	Store Staff / Cashier	10
5.	Non-Functional Requirements	12
6.	Success Metrics	13

1. Executive Summary

Borama Hardware is a multi-location hardware retail platform built for Borama, Somaliland. It operates in the style of Home Depot — a comprehensive, category-rich hardware store serving construction professionals and homeowners across the city.

The platform serves four user types across a customer-facing storefront, an in-store Point of Sale (POS) system, a back-office administration panel, and an inventory management portal. It is bilingual (English and Somali), supports dual-currency display (USD and Somaliland Shilling), operates offline via PWA, integrates mobile money payment confirmation, and uses Google Gemini AI for smart search, project estimation, inventory forecasting, and a hardware assistant chatbot.

Feature	Description
Bilingual UI	Full English and Somali support with per-user language preference
Dual Currency	SLSH (primary) and USD displayed on every price
Mobile Money	Zaad, eDahab, EVC Plus, Sahal — manual staff confirmation flow
PWA / Offline	Installable on Android, works offline for browsing and cash POS sales
AI-Powered	Semantic search, project estimator, inventory forecasting, chatbot
Multi-Location	Multiple Borama branches, each with independent stock and POS
Loyalty Program	Points-based rewards with Bronze, Silver, and Gold tiers

2. Problem Statement

Hardware retail in Borama currently operates with no digital infrastructure. The following problems exist today:

Problem	Impact
No product discovery	Customers cannot browse or search products before visiting a store.
No stock visibility	There is no way to check if an item is available across branches before travelling
Manual receipts	Cashiers write paper receipts. No digital records, no audit trail.
No loyalty program	Repeat customers receive no recognition or reward.
No management data	Store owners have no real-time sales, inventory, or staff performance data.
Language barrier	No Somali-language digital commerce tool exists for this market.
Currency confusion	USD and SLSH are both used; no unified pricing display.
Manual payment tracking	Mobile money confirmations are done verbally with no records.

3. Users & Roles

The platform has four distinct user roles, each with scoped access and responsibilities:

Role	Type	Primary Responsibility
Customer	External	Browse catalog, place orders, track order status, earn and redeem rewards
Cashier / Staff	Internal	Operate POS, verify mobile money payments, manage order pickup
Inventory Manager	Internal	Manage stock levels, receive stock, conduct stocktakes, view AI forecasts
Admin / Owner	Internal	Full platform access — products, staff, discounts, AI insights, reports

Access Control: All access is enforced at the database level using Supabase Row Level Security (RLS). Staff can only see data for their assigned location. Customers can only see their own orders and profile. Audit logs are visible to Admin and Owner only.

4. Feature Requirements

4.1 Customer-Facing Store

FR-C01 — Phone OTP Registration & Login

Customers register and log in using their Somaliland phone number (+252). An OTP is sent via SMS. Phone format is validated against known Somaliland prefixes (063, 065, 068, 061, 062, 066). First-time users complete a one-step onboarding screen to set their name and language preference. No email or password required.

FR-C02 — Bilingual Interface

All customer-facing text is available in English and Somali. A toggle in the header and account settings switches language immediately. Preference is stored per user profile and persisted across sessions.

FR-C03 — Dual Currency Display

All prices display in both Somaliland Shilling (SLSH) and US Dollars (USD). Primary currency is SLSH. Users can set their preferred primary currency in account settings. Conversion uses the exchange rate set by admin in the back-office.

FR-C04 — Product Browsing

Customers browse products through a hierarchical category system (13 top-level categories, multiple subcategories each) matching Home Depot's structure. Filters: price range, brand, in-stock status, subcategory. Sort: Featured, Price Low-High, Price High-Low, Newest. 24 products per page with pagination.

FR-C05 — Product Variants

Products in multiple sizes, colours, or specifications display a variant selector on the detail page. Selecting a variant updates the displayed price, SKU, and stock. Cannot add to cart without selecting all variant attributes first.

FR-C06 — Product Detail Page

Shows: image gallery (up to 8 images, swipeable on mobile), bilingual name and description, SKU, brand, dual-currency price, variant selector, per-location stock status, quantity selector, Add to Cart button, loyalty points to earn, Project Estimator CTA, and AI-powered related products.

FR-C07 — Semantic Search

Natural language search in English and Somali powered by Google Gemini AI embeddings. Searching 'fix a leaky pipe' returns plumbing products. Searching in Somali returns the same results. Falls back to cached full-text matching when offline.

FR-C08 — Shopping Cart

Persists across browser sessions and offline via IndexedDB. Customers can update quantities, remove items, apply discount codes, redeem loyalty points, and select a pickup location. Cart re-validates stock availability on checkout.

FR-C09 — Checkout

Single-page checkout with order summary, payment method selection (Zaad, eDahab, EVC Plus, Sahal, Cash on Pickup), mobile money phone number input, and optional order notes.

FR-C10 — Payment — Mobile Money (Manual Confirmation)

After placing an order, the customer sees the store's merchant account number, the exact amount to send, and the order number as reference. Customer sends money independently from their own phone. Customer taps 'I Have Sent the Payment' to notify staff. Staff confirms receipt on the Staff Payment Verification screen. Customer receives real-time notification and SMS when confirmed.

FR-C11 — Payment — Cash on Pickup

Customer selects Cash on Pickup. Order is confirmed immediately. Payment is collected at the store counter when the customer collects their order. No payment verification step required.

FR-C12 — Order Confirmation & Pickup Code

After checkout: order number, a 6-character alphanumeric pickup code (large, prominent), payment instructions (if mobile money), and a WhatsApp share button to forward order details to someone collecting on the customer's behalf.

FR-C13 — Order Tracking

Customers view all orders in My Orders. Each order shows status (Pending, Confirmed, Ready for Pickup, Completed, Cancelled), items, total, payment status. Order detail shows a status timeline and pickup code. Subscribes to Supabase Realtime for live status updates without page refresh.

FR-C14 — Order Cancellation

Customers can cancel an order while it is in Pending or Pending Verification status and payment has not yet been confirmed. Cancellation requires selecting a reason. Inventory reservation is released immediately. If payment was already confirmed, cancellation routes to admin.

FR-C15 — Returns

Customers initiate a return from a completed order by selecting items, specifying quantities, choosing a reason (Wrong item, Defective, Changed mind, Other), and selecting refund method (Cash at store, Mobile money, Store credit as loyalty points). Staff processes the return.

FR-C16 — Loyalty Program

Every customer is automatically enrolled on account creation with a unique digital loyalty card number. Earn 1 point per SL 1,000 spent. Three tiers: Bronze (0 pts), Silver (500 pts, 3% discount), Gold (2,000 pts, 7% discount). Points redeemable at checkout. Tier upgrades automatically based on lifetime points accumulated.

FR-C17 — AI Project Estimator

Customers describe a project in natural language ('I want to tile a 3m x 4m bathroom floor'). Gemini AI calculates required materials from the product catalog (tiles, grout, adhesive, spacers) with exact quantities and total cost. 'Add All to Cart' adds all in-stock items.

FR-C18 — Hardware Assistant Chatbot

Floating chat widget on the customer store. Customers ask hardware questions in English or Somali and receive AI-powered answers with product recommendations linked to the catalog.

FR-C19 — Store Locations

Page listing all Borama branch locations with address, phone number, and opening hours (including Friday prayer break). 'Get Directions' opens Google Maps.

FR-C20 — Account Settings

Customers can update their full name, preferred language, preferred currency, and change their phone number (OTP verified on the new number before the change takes effect).

FR-C21 — PWA & Offline Mode

Customer store is installable as a PWA on Android. When offline: cached products displayed with a banner notice, cart preserved, orders queued for submission on reconnect, checkout blocked until online.

4.2 Admin / Store Owner

FR-A01 — Admin Dashboard

Overview page with today/week/month stat cards (revenue, orders, new customers, low stock alerts). Charts: revenue trend, top products, orders by location, orders by payment method, return rate. Exchange rate staleness alert shown when rate has not been updated in more than 24 hours.

FR-A02 — Product Management

Full CRUD for products. Fields: name (EN + Somali), SKU, category, brand, unit, price in USD and SLSH (auto-converted), cost price, description (EN + Somali, AI-generatable), images (up to 8), dimensions, tags, featured/active toggles. Bulk CSV import with downloadable templates. Soft delete. Audit trail on all changes.

FR-A03 — Product Variants Management

Admin defines variant attribute groups (e.g., Size: M4, M6, M8; Length: 25mm, 50mm). System auto-generates all attribute combinations. Each variant has its own SKU, price, image, and can be individually activated or deactivated.

FR-A04 — AI Product Descriptions

One-click AI generation of bilingual product descriptions from product name, category, and brand. Generated text pre-fills the form for staff review and editing before saving.

FR-A05 — Category Management

Hierarchical category tree with add, edit, reorder (sort order), icon/image upload, and active/inactive toggle. Categories with existing products cannot be deleted.

FR-A06 — Order Management

Full order list with filters (location, status, payment method, date range). Order detail with admin actions: update status, confirm payment (enter reference number), generate PDF receipt, add internal note, cancel order, initiate return.

FR-A07 — Staff Management

Add staff by phone number, assign role (Cashier, Inventory Manager, Admin), and assign to a location. Edit role or location assignment. Deactivate staff accounts. Staff can only access data for their assigned location.

FR-A08 — Location Management

Add and edit Borama store branches with name (EN + Somali), address, phone, opening hours per day (with configurable Friday prayer break), and active/inactive toggle.

FR-A09 — Discount Management

Create discount codes: percentage off, fixed SLISH amount, fixed USD amount. Set minimum order, max total uses, max uses per customer, applicability scope (all products / category / specific products), and validity date range with automatic activation and expiry.

FR-A10 — Loyalty Program Admin

Configure tier thresholds, discount percentages, and tier names (bilingual). Manual point adjustment for any customer with a required reason (creates a loyalty transaction record).

FR-A11 — Exchange Rate Management

Update USD to SLISH exchange rate. Rate history log. Alert on admin dashboard when rate is stale. All SLISH prices on the customer store recalculate dynamically based on the current rate.

FR-A12 — Payment Settings

Configure merchant account numbers per mobile money provider (Zaad, eDahab, EVC Plus, Sahal) per location. Set custom bilingual payment instructions shown on the customer checkout confirmation screen.

FR-A13 — Banner Management

Upload and schedule homepage banners with bilingual title and CTA. Set active date range (for Eid sales, seasonal promotions). Target all locations or a specific branch. Drag-to-reorder.

FR-A14 — Customer Management

View all customers with total orders, loyalty tier, and total spend. View individual profile, order history, loyalty transactions, return history. Deactivate customer accounts.

FR-A15 — Returns Management

View all return requests across locations. Process returns: approve (restocks inventory, executes refund) or reject (with reason sent to customer via SMS). View return history.

FR-A16 — Audit Log

Immutable log of all admin data changes (product edits, price changes, discount creation, inventory adjustments). Shows who changed what, from what value, to what value, and when. Filterable by date, user, and table. Exportable to CSV.

FR-A17 — AI Insights Dashboard

View Gemini-generated inventory forecasts: predicted stockout dates, recommended reorder quantities, confidence scores, and AI reasoning per product. Trigger forecast regeneration on demand.

FR-A18 — Analytics & Reports

Revenue by date range and location. Top selling products. Order counts by status. Return rate trends. POS session summaries. Staff performance (sales per cashier). All reports exportable to CSV.

4.3 Inventory Manager

FR-I01 — Inventory Dashboard

Summary cards showing total SKUs, items in stock, low stock count, and out-of-stock count. Live unresolved alert feed. Real-time updates via Supabase Realtime.

FR-I02 — Stock Levels View

Per-location stock table with product, variant, quantity on hand, quantity reserved, quantity available (on hand minus reserved), threshold, aisle/shelf location, last restocked date, and status badge. Search by product name or SKU. Quick inline stock adjustment. Export to CSV.

FR-I03 — Receive Stock

Form to log incoming stock deliveries. Select location, add product/variant rows with quantity received and optional unit cost. On submit: increases stock, creates stock movement records (type: receive), resolves active alerts for restocked items, updates last restocked date.

FR-I04 — Stock Transfers

Transfer stock between Borama branch locations. Validates source has sufficient stock. Creates transfer-in and transfer-out stock movement records for both locations simultaneously.

FR-I05 — Inventory Alerts

List of all unresolved low-stock and out-of-stock alerts sorted by urgency (out-of-stock first). 'Mark Resolved' action. Quick link to Receive Stock for each alerted product.

FR-I06 — Stocktake

Initiate a physical inventory count for a location. System generates a count sheet with all products/variants and their current system quantities. Staff enter actual counted quantities (scannable by barcode). Discrepancies highlighted. Completed stocktake submitted for admin approval. Approved stocktake adjusts inventory to actual counts with full audit trail.

FR-I07 — AI Inventory Forecasts

Gemini-generated forecast table: predicted stockout date, recommended reorder quantity, confidence score, and AI reasoning per product per location. Sorted by urgency. Regenerate forecasts on demand from 30/60/90-day sales data.

FR-I08 — Inventory Reports

Four report types: Stock Valuation (quantity x cost price), Stock Movement (all ins/outs in a date range), Slow Moving Products (zero sales in last 30/60/90 days), and Damaged/Lost Goods (negative adjustments). All reports exportable to CSV.

4.4 Store Staff / Cashier

FR-S01 — POS Interface

Full-screen dual-column POS: product search and category grid on the left; cart and payment on the right. Touch-friendly large tiles. Category quick-filter buttons. Click or scan to add products. Variant modal appears for products with multiple options.

FR-S02 — POS Barcode Scanning

Barcode scanner input detected by 200ms keystroke timing + Enter key. Scanned SKU looked up instantly and product added to cart. 'Product not found' toast if SKU not in catalog.

FR-S03 — POS Cart

Cart shows product, variant, adjustable quantity, unit price, line total, and remove button. Discount code field. Optional customer phone lookup to link the sale to a loyalty account for automatic points accumulation.

FR-S04 — POS Cash Payment

Enter cash amount received from customer. System calculates and displays change automatically.

FR-S05 — POS Mobile Money

Staff selects mobile money provider, order is created, customer is advised to send to the store's merchant number. Staff verifies the incoming transfer on the store phone, then confirms it in the system with optional transaction reference entry.

FR-S06 — POS Split Payment

Split a transaction across two or more payment methods (e.g., part cash, part Zaad). Cash splits calculate change. Each mobile money split creates a separate verification record.

FR-S07 — POS Void Transaction

Void a completed POS transaction within the current session (within 60 minutes of sale). Restores inventory. Generates a void receipt with VOID watermark. Updates session refund totals.

FR-S08 — POS Receipt

Thermal-style 80mm receipt with store details, cashier name, order number with barcode, items table, totals, payment method, cash change, and loyalty points earned. Bilingual thank-you message. Print via browser print dialog. Download as PDF option.

FR-S09 — Barcode Label Printing

Print SKU barcode labels (50mm x 30mm) for products from the product page. Each label shows product name, variant, SKU barcode, and SLSP price. Set quantity to print.

FR-S10 — POS Session Management

Open session with starting cash count (SLSH + USD). Close session with end-of-day summary: total sales, breakdown by payment method, refund totals, cash variance indicator. Each cashier accesses only their own session data.

FR-S11 — POS PIN Lock

After 5 minutes of inactivity, POS locks and requires the cashier's 4-digit PIN to unlock (set during first login). Three failed attempts requires full phone OTP re-authentication.

FR-S12 — Parked Transactions

Cashier can park (hold) the current cart to serve a different customer, then recall the parked cart when ready to resume.

FR-S13 — POS Offline Mode

When offline, POS shows an OFFLINE badge. Cash payments continue. Sales are queued to IndexedDB and synced on reconnect. Mobile money options are disabled while offline.

FR-S14 — Payment Verification

Real-time list of all orders with pending mobile money confirmation for this location. Staff confirms each by entering the transaction reference from their phone. Triggers SMS to customer on confirmation.

FR-S15 — Order Pickup Management

Real-time list of confirmed online orders for this location. Mark order as 'Ready for Pickup' (triggers SMS to customer with pickup code). Verify pickup code when customer arrives (order marked Completed on successful match).

FR-S16 — Returns Processing

View return requests for this location. Approve (restocks inventory, processes refund by the customer's chosen method) or reject (with reason sent to customer via SMS).

5. Non-Functional Requirements

NFR-01 — Performance

Homepage loads in under 3 seconds on a 4G mobile connection. Product listing renders within 2 seconds. POS interface is interactive within 1.5 seconds of session open.

NFR-02 — Offline Support

Customer store PWA works offline for browsing (cached products) and cart management. POS continues processing cash sales offline with order queue. All queued orders sync automatically on reconnect via Background Sync API.

NFR-03 — Accessibility

WCAG 2.1 AA compliance. All interactive elements have ARIA labels. Minimum colour contrast ratio of 4.5:1. Full keyboard navigation support. Motion animations respect prefers-reduced-motion.

NFR-04 — Security

All routes protected by role-based middleware. Supabase Row Level Security enforced on every table. API rate limiting: OTP (3/hr/phone), search (30/min/IP), AI chat (20/hr/user), order creation (5/min/user). POS PIN prevents unauthorised terminal access.

NFR-05 — Reliability

Payment confirmation does not rely on third-party API availability. Mobile money uses manual staff confirmation — no integration dependency, no single point of failure. Failed SMS notifications never block the main order flow.

NFR-06 — Scalability

Multi-location architecture from day one. Each location has independent inventory, POS sessions, and staff assignment. Adding a new branch requires only inserting a record in the locations table — no code changes needed.

NFR-07 — Audit & Compliance

All admin data changes are written to the audit_log table with before/after values and the performing user's identity. POS sessions track all transactions. Stock movements record every inventory change with the user who performed it.

NFR-08 — Localisation

Full Somali and English support across all user interfaces. Phone number validation against Somaliland prefix patterns. Currency formatting: SLSH with no decimal places, USD with 2 decimal places. Friday prayer break configurable in store opening hours.

6. Success Metrics

The following metrics will be used to measure the success of the platform in the first 6 months after launch:

Metric	Target	Timeframe
Customer online order rate	20% of total sales	3 months
POS transaction speed	Under 90 seconds per sale	Day 1
Inventory accuracy	Less than 2% discrepancy	Monthly stocktake
Low stock alert response time	Under 4 hours to reorder	Ongoing
Customer loyalty enrollment	60% of customers	6 months
Search relevance	Under 5% no-results rate	Ongoing
Offline availability	100% cash POS uptime	Ongoing
Staff payment confirmation time	Under 10 minutes	Ongoing

Post-Launch Roadmap (v1.1+):

- Bulk / contractor ordering portal
- Supplier / vendor portal
- Mobile app (React Native)
- WhatsApp Business API integration for order notifications
- Advanced analytics with AI-powered business insights

Borama Hardware Store
AI-Powered Retail Platform
Borama, Somaliland

This document is confidential and intended solely for
the development team. Version 1.0 — June 2025